

CliniScan Lung-Abnormality Detection on Chest X-rays using AI

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1. Project Description

CliniScan is a deep learning-based medical imaging project designed to detect abnormalities in chest X-ray images. The system analyzes radiology images and identifies potential diseases by drawing bounding boxes around abnormal regions. This project aims to assist radiologists and healthcare professionals by providing automated detection of chest abnormalities.

The model used in this project is YOLOv8 (You Only Look Once), a state-of-the-art object detection algorithm capable of detecting multiple abnormalities in a single X-ray image. The system processes chest X-ray images, predicts disease regions, and provides visual outputs highlighting abnormal areas.

2.Dataset Used

The dataset used in this project is the VinBigData Chest X-ray Abnormalities Detection dataset. This dataset contains thousands of annotated chest X-ray images used for detecting various thoracic diseases.

Dataset components include:

- Train Images: X-ray images used to train the detection model.
- Test Images: Images used for evaluation.
- train.csv: Contains bounding box annotations and disease labels.
- sample_submission.csv: Example format for predictions.

The dataset includes abnormalities such as:

Aortic Enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD, Infiltration, Lung Opacity, Nodule, Pleural Effusion, Pleural Thickening, Pneumothorax, Pulmonary Fibrosis, and others.

3.Environment Setup

The following tools and technologies were used to implement the project:

Programming Language: Python

Deep Learning Framework: PyTorch

Object Detection Framework: YOLOv8 (Ultralytics)

Libraries Used:

- torch
- torchvision
- pandas
- numpy
- scikit-learn
- ultralytics
- Pillow

Hardware Used:

- CPU based training (Laptop)
- AMD Ryzen processor

4.Data Exploration

Data exploration was performed to understand the structure of the dataset and the distribution of different disease classes. The train.csv file was analyzed to identify the number of images, disease types, and bounding box annotations.

Key observations:

- Large number of chest X-ray images.
- Multiple abnormalities can appear in one image.
- Class imbalance exists where some diseases appear less frequently.

5.Data Processing

The dataset images were originally provided in DICOM format. These images were converted into JPG format to simplify processing and compatibility with the YOLO model.

Processing steps included:

1. Convert DICOM images to JPG format.
2. Extract bounding box coordinates from train.csv.
3. Convert bounding boxes to YOLO format (center_x, center_y, width, height).
4. Split dataset into training and validation sets.
5. Organize images and labels into YOLO folder structure.

6. Proposed System

The proposed system uses the YOLOv8 object detection model to identify abnormalities in chest X-ray images. The model takes an X-ray image as input and outputs bounding boxes highlighting abnormal regions.

System workflow:

1. Input chest X-ray image.
2. Image preprocessing and resizing.
3. Feature extraction using deep neural networks.
4. Object detection using YOLOv8.
5. Output prediction with bounding boxes and disease labels.

7. Training Pipeline

The training process involves the following steps:

1. Load dataset and annotations.
2. Configure YOLOv8 model.
3. Train model using training images.
4. Validate model using validation dataset.
5. Save best performing model weights.

Training parameters used:

- Model: YOLOv8n
- Epochs: 20
- Image Size: 512
- Batch Size: 16

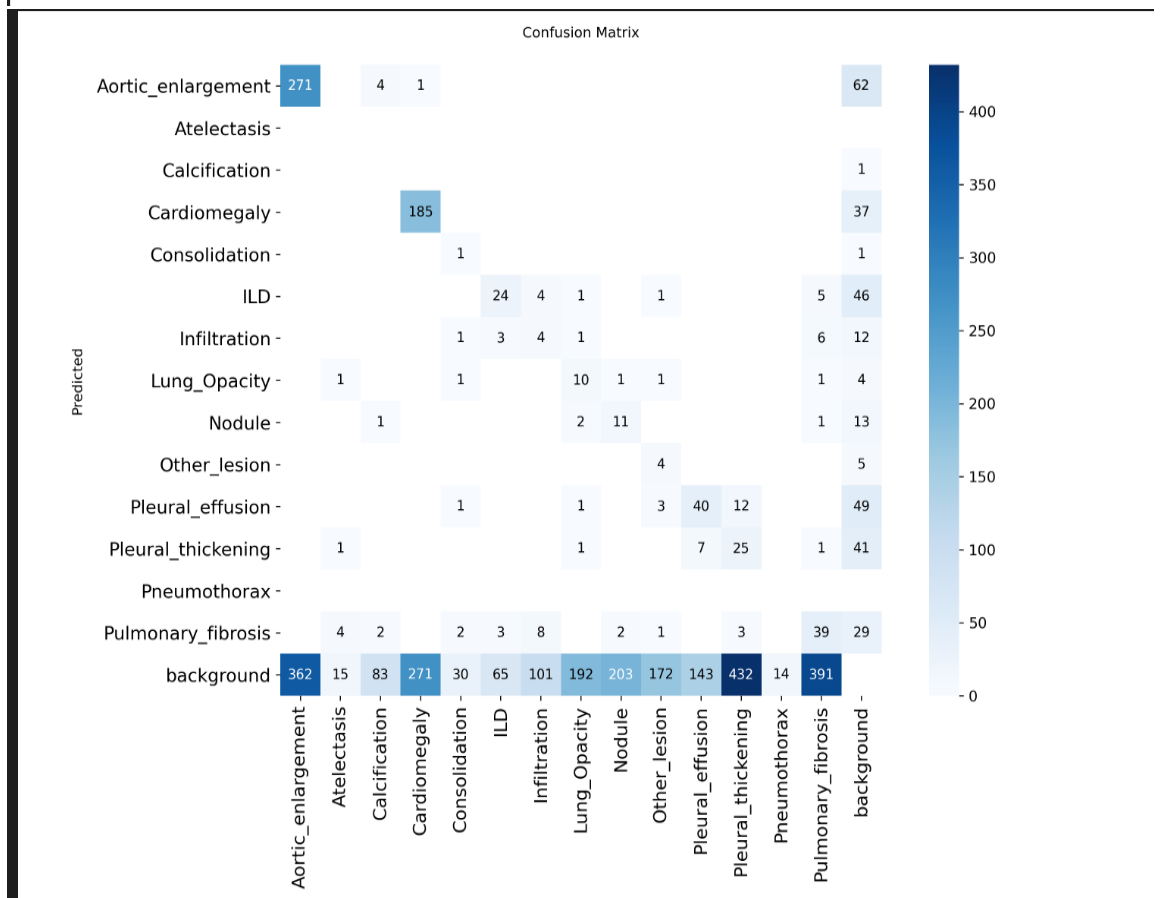
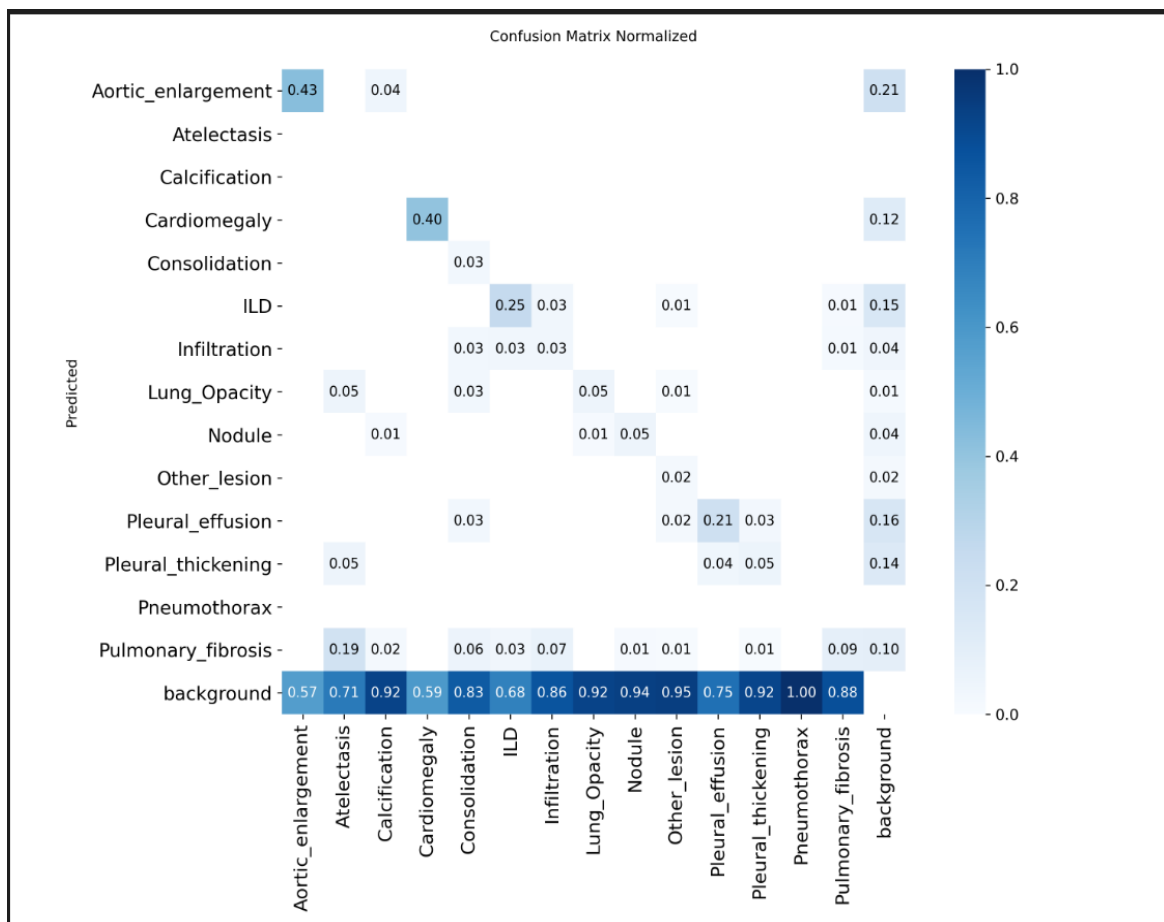
8.Results

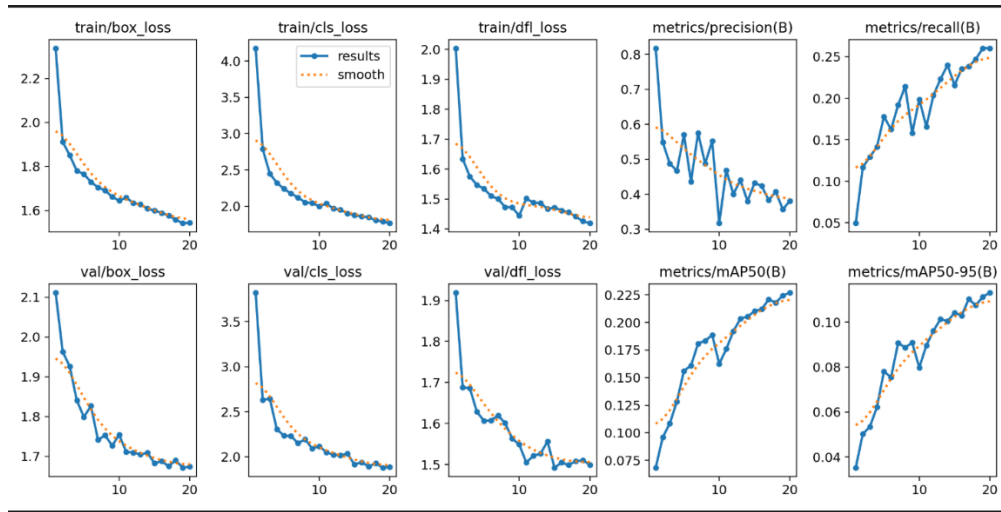
The trained YOLOv8 model successfully detected multiple abnormalities in chest X-ray images. The model produced bounding boxes around abnormal regions and predicted corresponding disease labels.

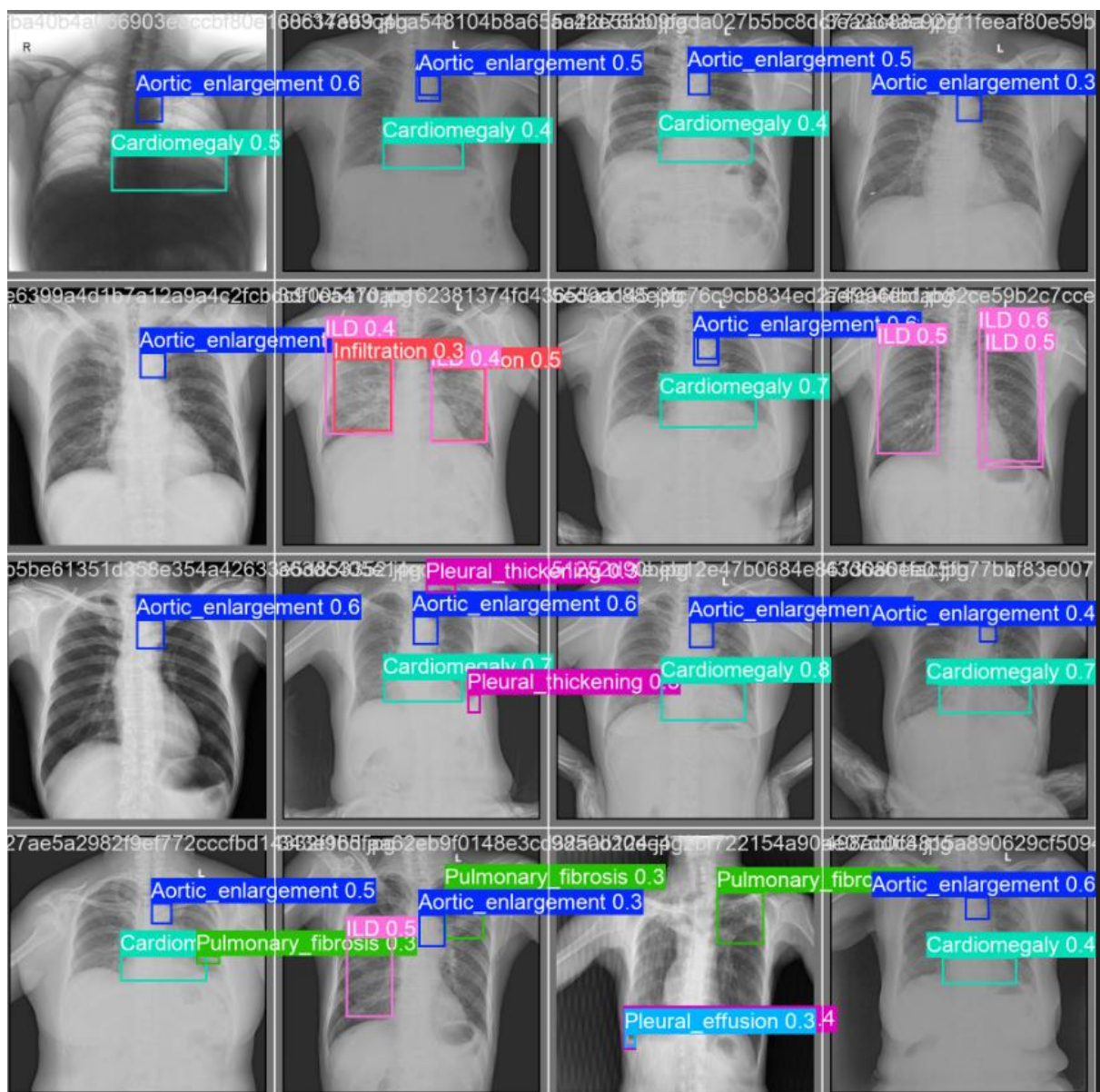
Evaluation metrics include:

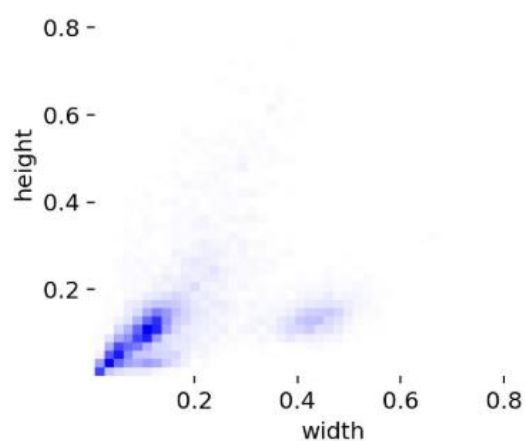
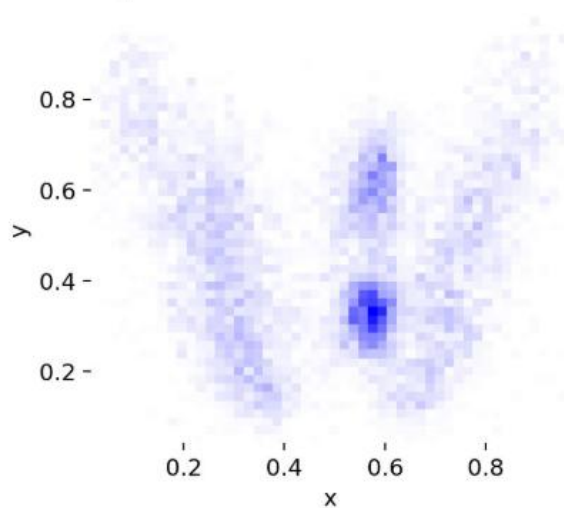
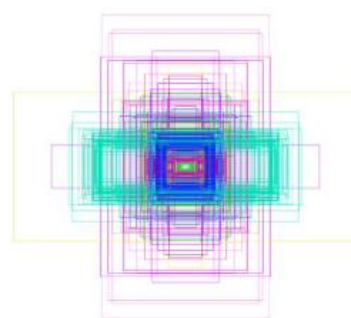
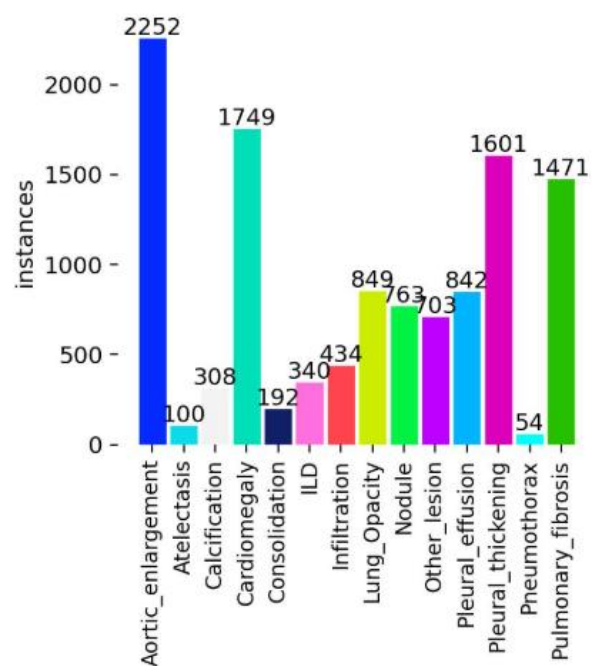
- Precision
- Recall
- F1 Score
- Mean Average Precision (mAP)

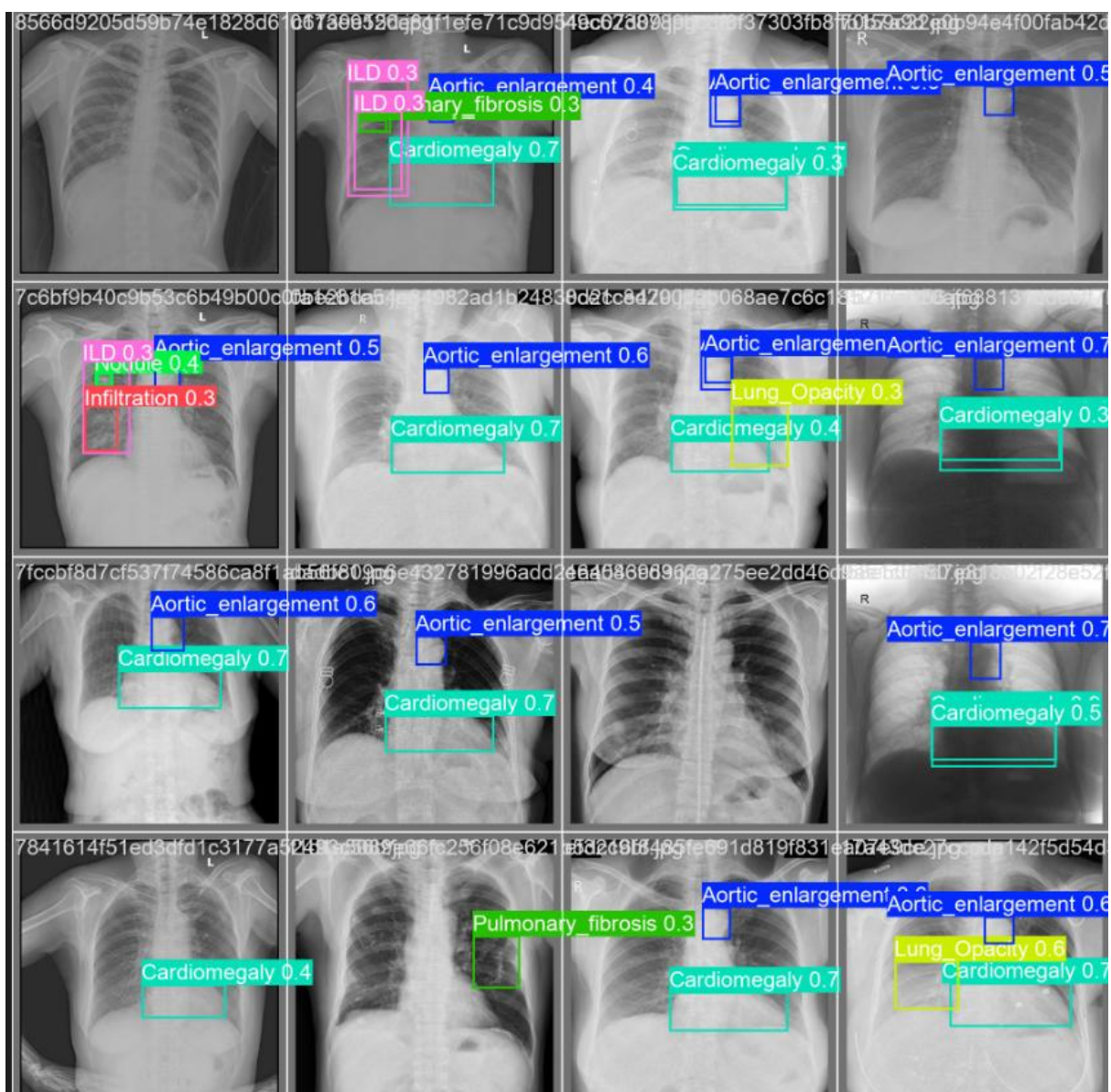
Sample detection results demonstrate that the model can correctly identify abnormal regions in several chest X-ray images.











9. Conclusion

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This project demonstrates the application of deep learning for automated detection of chest abnormalities using X-ray images. The YOLOv8 model proved effective in detecting disease regions and highlighting abnormal areas.

The system can assist radiologists by providing quick and automated analysis of chest X-ray images. Future improvements could include training on larger datasets, using GPU acceleration, and integrating the model into a clinical decision support system.